# Category Theory for Mathematical Physics — Exercise Sheet 12

*Morita equivalence and the Drinfeld centre*

**Variant: Questions only**

Conventions as in Lecture 12: a half-braiding is $\gamma_X : X \otimes Z \to Z \otimes X$, and the braiding of $\mathcal{Z}(C)$ is $c_{(Z,\gamma),(Z',\gamma')} = \gamma'_Z$ (Theorem 12.7). The toric code's anyons are $1 = (0,1)$, $e = (0,\chi_-)$, $m = (1,1)$, $\varepsilon = (1,\chi_-)$ in $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$ (Theorem 12.8), with $\mathbb{Z}_2 = \{0,1\}$ written additively and $\chi_-(h) = (-1)^h$; here $e$ is an anyon, and the identity of $S_3$ is written 1. $\zeta = e^{2\pi i/3}$, $\varphi = (1+\sqrt{5})/2$, and $\omega$ is a 3-cocycle (Exercise 6). S, T and the central charge $c$ are those of Lecture 10.

**Exercise 1 (The S-matrix of the toric code from half-braidings).** (i) Draw the condition of Definition 12.6 for a half-braiding on $Z = \delta_g$ in $\mathrm{Vec}_{\mathbb{Z}_2}$, with $X = \delta_h$ and $Y = \delta_k$, as in Equation (12.1). Writing $\gamma_{\delta_h} = \gamma(h)\,\mathrm{id}$, show that $\gamma$ is a character of $\mathbb{Z}_2$, and recover the four simple objects $(g,\chi)$ of Theorem 12.8.

(ii) Draw $c_{m,e}$ as in Equation (12.2), and show that the braiding is $c_{X,Y} = \chi_Y(g_X)$. Write the table of $c_{X,Y}$ and of the double braidings $c_{Y,X}c_{X,Y}$ for $X, Y \in \{1, e, m, \varepsilon\}$.

(iii) Using that every $d$ is 1 and every anyon is its own dual, show that $\tilde{S}_{XY}$ is the double braiding, and compare $S = \tilde{S}/2$ with Example 10.14.

(iv) Compute $T$ and the central charge.

**Exercise 2 (The double $D(G)$, and $G = \mathbb{Z}_3$).** (i) Show that $D(G)$ (Theorem 12.9) is associative, with unit $\sum_g \delta_g \otimes 1$, and has dimension $|G|^2$.

(ii) Let $G$ be abelian. Show that $D(G)$ is commutative, and that for $a \in G$ and a character $\chi$ of $G$, $\rho_{a,\chi}(\delta_g \otimes h) = \delta_{a,g}\,\chi(h)$ is an algebra map $D(G) \to \mathbb{C}$.

(iii) For $G = \mathbb{Z}_3$, list the nine irreducible representations and their spins $\theta = \chi(a)$, and compare with Theorem 12.8.

(iv) Label the simple objects of $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_3})$ by $(a,b)$, with flux $a$ and character $\chi_b(h) = \zeta^{bh}$. Using Definition 10.2 and the braiding of Theorem 12.8, show that $S_{(a,b),(a',b')} = \zeta^{-(ab'+a'b)}/3$. For invertible $S$ and $T$, $(ST)^3 = S^2$ is equivalent to $STS = T^{-1}ST^{-1}$: check the entry at $X = (0,1)$, $Y = (1,0)$, and show that it fails there for the complex conjugate $\bar{S}$ with the same $T$.

**Exercise 3 (The eight anyons of $\mathcal{Z}(\mathrm{Vec}_{S_3})$).** (i) List the conjugacy classes of $S_3$ and their centralisers, and the simple objects $([g],\pi)$ of Theorem 12.9.

(ii) The dimension of $([g],\pi)$ is $|[g]|\dim\pi$. Check that the squares add up to $|S_3|^2$.

(iii) Compute the spins $\theta = \chi_\pi(g)/\dim\pi$.

**Exercise 4 ($\mathrm{Rep}(\mathbb{Z}_2)$, and the symmetry $e \leftrightarrow m$).** (i) Following the proof of Theorem 12.2(ii), find the simple module endofunctors $(V,s)$ of $\mathrm{Vect}_{\mathrm{fd}}$ over $\mathrm{Vec}_{\mathbb{Z}_2}$ and their composition, and show that $(\mathrm{Vec}_{\mathbb{Z}_2})^*_{\mathrm{Vect}_{\mathrm{fd}}} \simeq \mathrm{Rep}(\mathbb{Z}_2) \simeq \mathrm{Vec}_{\mathbb{Z}_2}$.

(ii) Show that the exchange $e \leftrightarrow m$ (fixing 1 and $\varepsilon$) preserves the fusion rules, the spins and the double braidings of the toric code. Does it preserve each $c_{X,Y}$?

(iii) Write each anyon as $X = (g_X, b_X) \in \mathbb{Z}_2 \times \mathbb{Z}_2$ with $\chi = \chi_-^{b_X}$; the toric code has trivial associators. Let $F$ be the exchange $(g,b) \mapsto (b,g)$ on simple objects, extended linearly, with $J_{X,Y} = (-1)^{g_X b_Y}$ and $J_0 = 1$. Show that $(F, J, J_0)$ is a monoidal functor (Definition 3.10) and that it is *braided*, $F(c_{X,Y}) \circ J_{X,Y} = J_{Y,X} \circ c_{FX,FY}$: the exchange is a braided autoequivalence.

**Exercise 5 (Centres of modular categories).** You may use that on $C \boxtimes \mathcal{D}$ the braiding, the dimensions and the spins are taken factor by factor, $d_{a\boxtimes b} = d_a d_b$ and $\theta_{a\boxtimes b} = \theta_a \theta_b$, and that $C^{\mathrm{rev}}$ has the spins $\bar{\theta}_a$.

(i) By Theorem 12.10(iii), $\mathcal{Z}(\mathrm{Fib}) \simeq \mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$. List its simple objects with their dimensions and spins, and check $\dim \mathcal{Z}(\mathrm{Fib}) = (1+\varphi^2)^2$.

(ii) Show that for any unitary modular $C$ the Gauss sum of $C \boxtimes C^{\mathrm{rev}}$ is $\dim C$, so $c = 0$.

(iii) How many simple objects does $\mathcal{Z}(\mathrm{SU}(2)_k)$ have, and what is its dimension?

**Exercise 6 ⋆.** *The double semion.* Let $C = \mathrm{Vec}^{\omega}_{\mathbb{Z}_2}$, $\mathbb{Z}_2 = \{0,1\}$, with $\omega(1,1,1) = -1$, the other values 1.

(i) For a half-braiding on $\delta_g$, write $\gamma_{\delta_h} = \gamma(h)\,\mathrm{id}$ with $\gamma(0) = 1$. Show that Definition 12.6 becomes

$$\gamma(h+k) = \frac{\omega(g,h,k)\,\omega(h,k,g)}{\omega(h,g,k)}\,\gamma(h)\gamma(k).$$

(ii) Solve it: find the four simple objects of $\mathcal{Z}(C)$ and their spins.

(iii) Conclude with Theorem 12.11 that $\mathrm{Vec}_{\mathbb{Z}_2}$ and $C$ are not Morita equivalent.